

AI Literacy

POD 4 - Generative AI Tools for Everybody

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Today, we are exploring generative AI tools that can be adapted for our unique needs. Large language models, or LLMs, like ChatGPT, Gemini, and Claude, are versatile in that they can handle various tasks, from answering questions to summarising documents. But what if we want these tools to be even more specific—to handle the nuances of a specialised domain? This is where tailoring an LLM can make a real difference, and it is actually really easy to do. In this session, we will cover three techniques to personalise an LLM: prompt engineering, fine-tuning, and building a model from scratch. There is a fourth one called Retrieval Augmented Generation, or RAG but that is something the big tech companies do and is not something we ourselves can see or do directly ourselves.

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First, let us look at ways to tailor these models to meet specific needs. Using an “off-the-shelf” LLM like ChatGPT is a great starting point, but often, more specific adjustments are needed. The three main approaches are:

- Prompt Engineering, which is, crafting questions or commands to guide AI.
- Fine-Tuning, which is, adjusting the model with new data to specialise it for a certain area.
- Model Building, which is, building a model from scratch if the task is highly specific and requires unique training data.

Each of these methods has different use cases and resource requirements, so let's look at each in more detail. Prompt engineering is all about getting the best response by crafting the right input. It's a way to “talk” to the model in our everyday language



to guide its output without needing any computer programming. You might use prompt engineering to ensure the model gives you a brief summary instead of a lengthy response or to specify a certain tone in its output. For example, adjusting the prompt's “temperature” can make responses more creative (prompt by saying “set temperature = 1”) or more controlled (prompt by saying “set temperature = 0”). You can imagine this like fishing—prompt engineering is like choosing the right bait to get the specific answer you are after.

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With fine-tuning, we go a step further by actually modifying the model's parameters. This approach is useful when we want the model to specialise in a particular subject or style based on documents that we bring into the model. Imagine you need a model that understands intricate tax regulations. Fine-tuning allows us to train the model on a selected set of documents that we have, in order to “teach” it those specific details on top of what it already knows or has learned. An example here is the Revenue service in Ireland, where an LLM was fine-tuned to assist Revenue staff with tax-related queries. Staff members can simply ask the model a question, and it will pull relevant answers directly from a fine-tuned dataset. This makes the model an efficient assistant, saving time and effort in finding specific information. Let us look at how these techniques are being used in practical everyday settings. Companies like Microsoft are embedding LLMs in tools such as Copilot to help users draft emails or summarise reports. LinkedIn uses AI to enhance posts before you share them, while platforms like Zoom and Teams offer AI-generated meeting summaries and task assignments.

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These examples highlight how generative AI is becoming part of our everyday tools, simplifying tasks that would otherwise take up a lot of our time. To understand fine-tuning in practice, let's return to the case of Ireland's Revenue service. Revenue staff have to navigate hundreds of pages of technical tax manuals to understand client questions. By fine-tuning an LLM with these manuals, the model became a specialised assistant, capable of returning concise, relevant information on complex



tax issues. This project involved training the model on various tax-related documents, turning it into a powerful resource that can quickly answer detailed questions with accurate information, directly benefiting Revenue's staff and clients. But sometimes, even fine-tuning is not enough. In cases where a field has very unique terminology and knowledge—such as medicine or law—companies build specialised LLMs from scratch rather than fine-tuning an LLM from one of the big tech companies. These are models trained specifically on field-specific data and often require significant expertise and resources to gather data and do that training.

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For example, MedAlpaca is designed for medical professionals, and Harvey.AI is tuned for legal information. These specialised models cater to the unique needs of their industries, making them more accurate than general-purpose LLMs. Let us summarise the levels of customisation for LLMs, from the most basic to the most advanced:

- **LLM:** Basic model as-is.
- **LLM + Prompt Engineering:** Enhanced with guided input.
- **LLM + Fine-Tuning:** Custom data used to specialise the model.
- **Specialist LLM:** Built from scratch or extensively fine-tuned for a specific domain.

Each level increases the specificity and accuracy of responses, depending on the task requirements and available resources.

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In summary, LLMs and generative AI tools are becoming essential in both personal and professional contexts. With methods like prompt engineering, fine-tuning, and retrieval, we can adapt these tools to fit specialised needs, whether it is assisting with tax queries, writing reports, or providing medical guidance.



In the next POD, we will move to hands-on exercises where we will have a chance to experiment with prompt engineering and explore how generative AI can be tailored to support our own work.